

Network Segmentation Testing

CompTIA PenTest+ (PT0-003)

Domains:

- 1.0 Planning and Scoping
- 4.0 Post-Engagement Activities

Objectives:

- 1.5: Given a scenario, analyze the findings and recommend the appropriate remediation within a report.
- 4.2: Given a scenario, perform network attacks using the appropriate tools.

Video Overview

This video offers an in-depth look at network segmentation testing, a fundamental practice for any professional involved in cybersecurity or IT infrastructure. As you prepare for certifications or advance your career, understanding how to effectively segment networks and validate those controls becomes increasingly important. This material will not only help you analyze security findings and recommend appropriate remediation in various scenarios, but also equip you with the skills to prioritize and prepare for potential attacks—abilities directly applicable to real-world cybersecurity roles and essential for safeguarding organizational data.

Key Topics Covered

- **Network Segmentation Definition:** Dividing a network into smaller, isolated segments to prevent unauthorized lateral movement.
- **Purpose of Network Segmentation:** Reducing risk, protecting systems, and supporting compliance.
- **Regulatory Importance:** Compliance with PCI DSS and internal policies like ISO 27001.
- **PCI DSS Requirement 11.3.4:** Annual penetration tests and retesting after changes to segmentation controls.
- **Cardholder Data Environment (CDE):** Systems that store, process, or transmit cardholder data.
- **Segmentation Controls:** Firewalls, Access Control Lists (ACLs), Virtual Local Area Networks (VLANs), and routing configurations.

- **Basic Testing Tools:** Using `ping` and `Nmap` for discovering segmentation flaws.
- **Commercial Testing Tools:** Utilizing professional tools like Metasploit Pro for automated testing and reporting.

Student Notes Section